better guide;
- Grokking Deep Learning, by DeepMind's Andrew W. Trask, an excellent introduction, ideal to approach the world of deep learning;
- Deep Learning with Python, written by François Chollet, the author of Keras library: you'll learn how to use it to build image and text classification models, and generators;
- Deep Learning with JavaScript, in case you'd like to build models for the web that run in the browser, using *Tensorflow.js* – and written by its main authors.

There are, of course, many more great books out there – it's impossible to list them all here – but these are a good starting point.

16.5 Gradient Descent for Graph Embedding

So, now that we have discussed at length how gradient descent works, its (many) strengths and its flaws, you should be able to figure out how to apply it to our case study, a heuristic to find a straight-line drawing for graphs, with the minimum number of intersections between edges.

And your answer should be… that gradient descent can't really help: if we look at figures 16.7 to 16.12, it's clear that the cost function for "minimum number of intersections" is step-shaped, with large plateau regions (where the gradient is null) and sudden drops.

You might be wondering, then, why have we introduced gradient descent: some readers could guess what's the next step, but if you didn't, take a minute to mentally go over what we have